

# ROSHAN SHRESTHA, Ph.D.

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## PROFILE

Research Engineer with a PhD in Computational Biophysics, currently developing GenAI-powered document intelligence and automation systems at Materialise NV. Experienced in building LLM-based extraction pipelines, RAG prototypes, workflow orchestration, and structured metadata validation using Azure AI services, Microsoft Power Platform, and OpenAI/Anthropic APIs. Combines practical AI engineering with 4+ years of scientific computing, molecular simulation pipeline development, and technical coordination in EU-funded R&D projects.

## TECHNICAL SKILLS

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|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GenAI &amp; LLMs</b>       | OpenAI API, Anthropic API, Azure AI Document Intelligence, Copilot Studio, RAG, structured extraction, prompt engineering, tool/function calling |
| <b>Automation &amp; Cloud</b> | Power Automate, SharePoint, Dataverse, Azure DevOps, Docker, Git, CI/CD, workflow orchestration                                                  |
| <b>Programming</b>            | Python, Bash, SQL; working knowledge of C++, Fortran, TCL                                                                                        |
| <b>ML &amp; Data</b>          | PyTorch, scikit-learn, NumPy, Pandas, model evaluation, regression, classification, feature engineering                                          |
| <b>Scientific Computing</b>   | GROMACS, OpenMM, NAMD, Martini 3, molecular dynamics, enhanced sampling, HPC job scheduling                                                      |

## PROFESSIONAL EXPERIENCE

**Research Engineer — AI & Computational Modeling**  
Materialise NV | Leuven, Belgium

Sept 2025 – Present

### GenAI & Contract Automation

- Designed and implemented a contract lifecycle automation agent for legal workflows, covering document upload, metadata extraction, search, duplicate detection, status tracking, expiration alerts, and field accuracy reporting.
- Built an NDA metadata extraction pipeline integrating email ingestion, Azure Document Intelligence OCR, GPT 5 structured extraction, Teams adaptive-card review, and SharePoint metadata deposition.
- Engineered a constrained Copilot Studio extraction prompt for contract-party attribution, EU date normalization, affiliate handling, office resolution, and strict JSON output.
- Implemented contract-party deduplication, ingestion logging, failure monitoring, and daily Teams alerts using Dataverse and Power Automate.

### EU-Funded R&D Projects: Valid3D and PowerizeD

- Led technical writing and review of consortium deliverables, including architecture and data-exchange specifications, coordinating input from KU Leuven, IMEC, Sirris, and Maastricht University.
- Developed standardized simulation and data workflows connecting physics-based molecular models with industrial R&D pipelines for digital materials engineering.

**PhD Researcher — Computational Biophysics**  
Université Claude Bernard Lyon 1 | Lyon, France

2021 – 2025

- Developed and validated Martini 3 coarse-grained models for graphene, carbon nanotubes, and fullerene, enabling 2–3 orders-of-magnitude faster simulations than atomistic models while preserving key thermodynamic and structural properties.
- Built reproducible Python/Bash workflows for large-scale molecular dynamics simulation, trajectory analysis, parameter validation, and publication-quality data processing.
- Applied enhanced-sampling methods, including umbrella sampling and metadynamics, to quantify interfacial and thermodynamic properties relevant to material design.
- Published first-author and co-authored work in J. Chem. Theory Comput., Biomacromolecules, and J. Phys. Chem. B.

## SELECTED PROJECTS

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- **Multi-Model LLM Comparison Tool** — Python tool for comparing Anthropic and OpenAI models across response quality, latency, streaming behaviour, structured output, and tool-use capabilities.
- **RAG Learning Pipeline** — Retrieval-augmented generation prototype using BAAI/bge-small-en-v1.5 embeddings with Claude and local Ollama backends.
- **Contract Metadata Evaluation Framework** — Python-based comparison workflow for scoring LLM-extracted contract metadata against SharePoint ground truth using field-level accuracy, coverage, and normalized matching.
- **Martini3-Graphene** — Open-source coarse-grained force-field parameters for graphene, CNTs, and fullerene, validated against atomistic and experimental data ([GitHub](#)).

## SELECTED PUBLICATIONS

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- Shrestha, R., et al. *Martini 3 Coarse-Grained Models for Carbon Nanomaterials*. J. Chem. Theory Comput., 2025.
- Cambiaso, S., Shrestha, R., et al. *Martini 3 Coarse-Grained Model for Chitosan*. Biomacromolecules, 2025.
- Naz, Z., Shrestha, R., et al. *Interaction of Phthalates with Lipid Bilayer Membranes*. J. Phys. Chem. B, 2022.

## EDUCATION

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**PhD in Computational Biophysics** 2021 – 2025  
*Université Claude Bernard Lyon 1* | Lyon, France  
Thesis: *Coarse-grained modeling of carbon-based nanomaterials* | Supervisor: Prof. Luca Monticelli

**M.Sc. in Physics** 2019  
*Tribhuvan University* | Kathmandu, Nepal

## ADDITIONAL

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**Awards:** Best Poster — ACS ICAN 2019; BioExcel Winter School 2020; NanoTox 2024  
**Languages:** English (professional), Nepali (native), French (A2), Dutch (A2)  
**Interests:** Open-source tooling, photography, running, cycling